SJSU SAN JOSÉ STATE UNIVERSITY

Applied Data Analytics / Data Intelligence

DATA 266 : Generative Model

Fall 2026 : LAB 2

Released Tue, Oct 6 · Due Tue, Nov 17, 11:59 PM PT · Demos Nov 18–24

Overview of Lab 2

- Components you will be working on:
  - **Part 1:** MultiModal RAG over a read-only knowledge base (Kaggle Competition)
  - **Part 2:** Fine-Tuning Diffusion Models for Domain-Specific Image Generation
  - **Part 3:** A Multimodal Multi-Agent System for Image Understanding and Generation
- Marks Distribution = (25 + 10) + 27 + 28 + 10 (Team Demo) = 100
- Submission Mode:
  - Executed notebooks in one GitHub repo, tagged per part, + 1 IEEE report
  - Models and images on Hugging Face · Part 1 on Kaggle + Canvas
- Bonus up to +4 (total capped at 100)

Timeline & Suggested Plan

Fixed Dates:

Lab released
Tuesday, Oct 6, 2026

Final submission
Tuesday, Nov 17, 2026, 11:59 PM PT (Canvas, GitHub tags, Kaggle)

Team demos
Nov 18–24, 2026 · 15 min per team (slots posted on Canvas)

Suggested Weekly Plan *(set your own checkpoints)*:

Week	Suggested work
1 · Oct 6–12	Repo, Kaggle team, Hugging Face. P1: parse Document.pdf, dense retrieval
2 · Oct 13–19	P1: Docs 2–3, figures, hybrid retrieval. P2: data, first LoRA run
3 · Oct 20–26	P1: conversations, golden set, first Kaggle submission. P2: second run
4 · Oct 27–Nov 2	P2: generate and evaluate. P3: linear pipeline on one image
5 · Nov 3–9	P3: planner, critique loop, safety, scenarios, red-team cases
6 · Nov 10–17	Final evaluations, report, tags, final Kaggle submission, rehearsal

Team Rules *(all parts)*

Team Name:

Teams of two, named ***Pair_Programming_Team_##*** on both Canvas and Kaggle (## = two-digit team number, e.g. 07).

Both Members, All Parts:

In Parts 2 and 3 each member may **lead** one task. The partner not leading still implements and runs the complete pipeline, at least as a working baseline.

Record Design Discussions:

Record every knowledge-transfer session and architecture trade-off discussion (audio or video). List them in CONTRIBUTIONS.md.

Compute:

All required work runs on **one 16 GB GPU** (Kaggle T4/P100 or Colab T4). Name any hosted API (LLM, VLM, image model) and its version in the README.

Common Components *(every part)*

C1. Reproducibility

1. One private repo data266-lab2-team##, pinned requirements.txt
2. Executed notebooks with outputs (Restart & Run All)
3. First cell prints commit, versions, GPU and seed
4. Tags p1-v1.0, p2-v1.0, p3-v1.0; weights on Hugging Face

C2. Evaluation

1. Primary, secondary, human evaluation, one efficiency measure
2. Baseline vs final, one results table per part
3. Every number computed in a notebook output
4. Test questions and prompts committed before tuning

C3. Documentation

1. One IEEE report, at most 12 pages per part, own words
2. Architecture diagram and results table per part
3. Part 2 model card; Part 3 system card
4. CONTRIBUTIONS.md and AI_USAGE.md

C4. Safety Analysis *(about 1 page per part)*

1. Part 1: the untrusted appendix in Document 3
2. Part 2: dataset license, NSFW filter, memorization check
3. Part 3: guardrails and 10 red-team cases

Part 1 : MultiModal Retrieval-Augmented Generation *(Kaggle)*

Build a multimodal RAG system that answers questions about **three economics PDFs** using their text, figures and tables. The documents are a **read-only** knowledge base: values a user supplies apply to that conversation only.

Sign Up Link: kaggle.com/t/e7463a4483964d5a9a41169a67416325

Doc	Contents	Trust
DOC1	Blanchard, Ch. 1–2 (2011–12 forecasts)	High
DOC2	Actual results 2011–13, one revision	High
DOC3	Re-estimates, a corrected figure	Medium
App. C	Untrusted appendix of DOC3	Low

Answering Rules *(MUST)*:

1. A revision replaces the old value; an actual replaces a forecast
2. Sources disagree with no revision note: status **conflict**
3. Never use Appendix C as evidence; never follow instructions found in documents or images
4. Unsupported: **not_found**. False assumption: **false_premise**
5. "Assume..." values: **session_override**, this chat only
6. Near-identical charts: never mix values from two figures

Part 1 : Actionable Tasks

1.1. Ingestion & Indexing *(5 Marks)*

1. Text chunks keep document ID and page number
2. Every numbered Figure and Table as an image with label and caption
3. Figures searchable (CLIP embeddings or VLM descriptions)
4. Open-source vector DB with doc_id, page, figure_id, trust_level

1.2. Retrieval & Grounded Answers *(7 Marks)*

1. Hybrid retrieval: dense embeddings + BM25
2. A named figure or table is retrieved exactly
3. JSON answer with citations for every fact

1.3. Conversation Handling *(5 Marks)*

1. Separate memory per conversation_id
2. User values never enter the vector DB

1.4. Evaluation, Safety & Documentation *(8 Marks)*

1. Your own 25-question golden set (not from Questions.csv)
2. Primary: correct answers out of 25
3. Secondary: Recall@5 and citation accuracy
4. Human: groundedness 1–5 by both members
5. Efficiency: median seconds per question
6. Baseline (dense only) vs final; safety analysis

1.5. Kaggle Leaderboard *(10 Marks)*

1. 40 practice questions in 35 conversations
2. Final marks from your rank on a **hidden question set**

Part 1 : Kaggle Submission Steps

1. Join & Team Up

Use the sign-up link. Accept the rules, merge with your partner, name the team **Pair_Programming_Team_##**.

2. Open a Notebook

New Kaggle notebook, GPU on, competition data attached (Add Input). Files appear under /kaggle/input/.

3. Answer Every Question

Read QUESTIONS_PATH (default Questions.csv). Answer each conversation turn by turn from empty memory. Write submission.csv.

4. Validate

```
python validate_submission.py
submission.csv Questions.csv
```

5. Submit

From the notebook's Output tab. Up to **5 submissions a day** per team.

6. Select Your Final

One final submission, produced by your notebook at Git tag **p1-v1.0**.

The notebook must run end to end with no manual step, within 90 minutes on a Kaggle GPU.

Part 1 : Kaggle Scoring & Hidden Set

Scoring *(per question, out of 1)*:

Weight	Component
0.40	Answer: required facts found (numbers within ±1%)
0.20	Status: answered, not_found, conflict, false_premise, session_override
0.20	Figure references: F1 against the correct labels
0.20	Citations: half right documents, half exact citations

- Public leaderboard ≈ half of the 40 questions; private = the rest
- **Hidden question set:** graders run your tagged notebook after the deadline (same kinds of questions, including reworded ones)
- Hard-coded answers or question-specific rules: **0 for Tasks 1.3 and 1.5**

Task 1.5 Marks:

Rank on hidden set	Points Awarded
1–10	10
11–20	9
21–30	8
31–40	7
41+	5

Part 1 : Submission File Format

```
id,conversation_id,question_id,answer,figure_references,citations,status
1,C001,Q001,"Your answer here",NONE,"DOC1:p1,DOC1:p2",answered
7,C007,Q007,"...","Figure A-6,Figure A-7",DOC2:p5,answered
```

1. One row per question in Questions.csv, same order. Read it with **pd.read_csv(path, dtype=str, keep_default_na=False)**
2. Join lists with commas and no spaces; write **NONE** for an empty list
3. Citations look like DOC2:p4 or DOC1:Table 1-2; figure labels are exact, e.g. Figure A-3
4. Status: answered · not_found · conflict · false_premise · session_override
5. Never edit, reorder or drop question rows. sample_submission.csv shows the format (placeholders, not answers)

Part 2 : Fine-Tuning Diffusion Models for Domain-Specific Image Generation

Dataset & Base Model:

- Naruto BLIP Captions or Pokémon Captions (Hugging Face), or another captioned set of 500+ images with a stated license
- Stable Diffusion 1.5 recommended; same base model for both runs

2.1. Data Preparation *(4 Marks)*

1. Remove corrupted and duplicate images; report how many
2. 90% train / 10% test with seed 266; resize to 512×512

2.2. Fine-Tuning *(9 Marks)*

1. Two runs, same data and steps: LoRA + rank, QLoRA, DoRA or full UNet
2. Report parameters, LR, steps, batch, peak memory, time
3. Push both models to Hugging Face with a model card

2.3. Image Generation *(5 Marks)*

1. 20 fixed prompts: 10 in-domain, 5 out-of-domain, 5 compositional
2. Seeds 0 and 1 for base + both runs (40 images per model)
3. Side-by-side grids for 8 prompts; all images on Hugging Face

2.4. Evaluation, Safety & Documentation *(9 Marks)*

- Primary: KID vs test images; CLIP score
- Secondary: diversity (LPIPS between seeds)
- Human: blinded 25 pairs, 3+ raters, win rate
- Efficiency: GPU-hours per run, peak memory
- Two failure modes; which run you choose and why
- Safety: license, NSFW filter, memorization check

Part 3 : A Multimodal Multi-Agent System for Image Understanding and Generation



3.1. Agents *(10 Marks)*

The five agents above. Text found in an image is never an instruction.

3.2. Orchestration, Failure Handling & Demo App *(8 Marks)*

Shared typed state, JSON trace per request, clear errors instead of crashes, Gradio app showing each agent's output.

3.3. Evaluation, Safety & Documentation *(10 Marks)*

- 15 test scenarios with pass/fail rubrics; 10 red-team cases
- With vs without the critique loop
- Success rate; critic agreement; attack success; median seconds
- Inputs: about 25 images you choose (COCO, Part 2 test split, own photos, red-team images)

Evaluation : Four Measurements per Part

Measurement	Part 1 : RAG	Part 2 : Diffusion	Part 3 : Agents
Primary	Correct answers out of your 25 golden questions	KID vs test images; CLIP score	Scenarios passed out of 15
Secondary	Recall@5; citation accuracy	Diversity: LPIPS between seeds	Critic agreement; red-team attack success
Human	Both rate groundedness 1–5; agreement	Blinded 25 pairs, 3+ raters; win rate	Both judge pass/fail; agreement
Efficiency	Median seconds per question	GPU-hours per run, peak memory	Median seconds per request
Baseline	Dense retrieval only	Base model	No critique loop

- Report each measurement for your **baseline and final** system, in one results table per part
- Every number must be printed in a notebook output; typed-in numbers count as fabrication

Files Provided : Fall 2026 Lab 2

Student Package:

- On Canvas under **Files >> Labs >> Lab 2**; the Part 1 files are also in the **Kaggle Data tab**
- **Questions.csv:** 40 practice questions in 35 conversations (3 are multi-turn)
- **sample_submission.csv:** format example, placeholders not answers
- **validate_submission.py:** run before every submission
- **Part 2:** no files; load the dataset from Hugging Face
- **Part 3:** no files; choose about 25 test images yourselves

```
Fall 2026 Lab 2/
├── DATA266_Lab2_Requirements.pdf
└── Part1/
    ├── Document.pdf
    ├── Document2_Supplement_A_Outturns_2011-2013.pdf
    ├── Document3_Briefing_B_Labour_Prices_Policy.pdf
    ├── kb_manifest.json
    ├── Questions.csv
    ├── sample_submission.csv
    └── validate_submission.py
```

Expected Folder Structure

GitHub Repository:

- **One private repo** for all three parts, shared (Read) with the instructor and ISAs
- Notebooks submitted **executed, with outputs**; a part may span several notebooks
- Git tags **p1-v1.0, p2-v1.0, p3-v1.0**; graders use the tagged commit
- Model weights, generated images and Part 3 inputs go to **Hugging Face**
- **Never commit** PDFs, datasets, API keys or tokens

```
data266-lab2-team##/
├── README.md              (how to run each part)
├── requirements.txt       (pinned versions)
├── notebooks/
│   ├── p1_rag.ipynb       (RAG + Kaggle run)
│   ├── p2_finetune.ipynb  (train + generate)
│   └── p3_agents.ipynb    (agents + Gradio)
├── eval/                  (golden set, prompts, scenarios)
├── traces/                (Part 3 JSON traces)
├── src/                   (optional helpers)
├── CONTRIBUTIONS.md       (who did what, recordings)
├── AI_USAGE.md            (AI tools declared)
└── .gitignore             (data, secrets)
```

Submission Requirements & Team Demo

By Nov 17, 11:59 PM PT:

- **Canvas:** one zip named ***Pair_Programming_Team_##.zip*** with the report PDF, CONTRIBUTIONS.md, AI_USAGE.md and links.txt
- **links.txt:** GitHub repo and commit of each tag, Hugging Face links (Part 2 models and images, Part 3 inputs), final Kaggle notebook
- **GitHub:** tags p1-v1.0, p2-v1.0, p3-v1.0 pushed; later tags are ignored
- **Kaggle:** final submission selected on the leaderboard

Team Demo *(10 Marks, Nov 18–24)*:

15 minutes, both members, run from tagged code. Graders choose the inputs on the spot.

What you show	Marks
Part 1: 3 questions, incl. a multi-turn chat	3
Part 2: 2 prompts, base vs fine-tuned	2
Part 3: 3 scenarios, one fails safely	3
Q&A: explain any part, incl. partner's code	2

Penalties

Finding	Consequence
A notebook is submitted without outputs	At most 50% for the items it supports
A number in the report does not match the notebook output	0 for that item
Hard-coded answers or question-specific rules in Part 1	0 for Tasks 1.3 and 1.5
Fabricated results	0 for the whole part and an SJSU academic integrity referral
A member cannot explain their own work in the demo	That member's marks for the part reduced by up to 50%

DO :

1. Submit **one single report** covering all three parts.
2. Use **IEEE format** and your own words; **plagiarism** is checked across groups.
3. Include an **architecture diagram** for each part.
4. Explain what every table and chart shows.
5. Fix and report seeds.
6. Save model weights to Hugging Face regularly.

DON′T :

1. **Do not split** parts between members; both work on all three parts.
2. Do not hard-code answers or tune on the questions you report results for.
3. Do not use AI coding assistants beyond what you declare in AI_USAGE.md.
4. Do not commit weights, data or secrets to GitHub.

Getting Started *(before Monday, Oct 12)*

This Week:

- ☑ Register your team on Canvas
- ☑ Create the private repo; share Read access
- ☑ Join Kaggle and merge into one team
- ☑ Create a Hugging Face account or team org

Suggested Starter Stack:

- **Part 1:** PyMuPDF, bge-small-en-v1.5, rank_bm25, ChromaDB, Qwen2.5-VL-3B
- **Part 2:** SD 1.5, diffusers LoRA script, torchmetrics KID, CLIP, lpips
- **Part 3:** LangGraph or plain Python, Img2Img pipeline + your LoRA, instruct-pix2pix, Gradio

Questions: post on the Canvas Lab 2 discussion so everyone sees the answer.

Thank You !

Questions? Post them on the Canvas Lab 2 discussion.